

Review

Catalysis: Key Technology for the Conversion of CO₂ into Fuels and Chemicals

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Abstract

The sustainability of the chemical industry requires replacing oil and natural gas with alternative raw materials and reducing CO₂ emissions from processes and utilities. In the particular case of petrochemicals, decarbonization is not easy, since carbon is an integral part of the products. Fossil carbon can be replaced with recycled carbon and renewable carbon, but it is the use of CO₂ as a raw material that will finally make it possible to close the carbon cycle in the chemical industry. The options available are discussed herein, highlighting recent breakthroughs in catalysis and identifying areas where further research is needed.

Keywords: CO₂ valorization; catalysis; fuels and chemicals; “power-to-X”; decarbonization

1. Introduction

The concentration of carbon dioxide in the atmosphere has been rising steadily since the Industrial Revolution (from 280 ppm in 1780 to 425 ppm in 2024), contributing to global warming and causing significant climate change. The burning of fossil fuels is responsible for the majority of anthropogenic CO₂ emissions, amounting to around 36 Gt/year. A large part is retained by the soil, plants and oceans, but more than 40% accumulates in the atmosphere, increasing the CO₂ concentration by 2 ppm/year [1]. In order to meet the 2015 Paris Agreement targets, the energy transition must be based on four pillars: renewable electricity; biofuels; hydrogen; and carbon dioxide capture, utilization and storage (CCUS) [2].

The chemical industry is one of the sectors of the economy where decarbonization is considered most difficult, especially in the case of petrochemicals, since carbon is part of the products. In this case, fossil carbon can be replaced with renewable carbon (from biomass) and recycled carbon (namely by recycling plastics), but carbon dioxide can also be used as a raw material. The use of CO₂ as a carbon source for the synthesis of chemical products has a long history dating back to the 1970s [3,4], but the topic has now gained momentum by becoming one of the most relevant technologies for achieving carbon neutrality. In particular, it will be shown herein that transportation fuels, as well as the “base chemicals” (or primary products) on which the whole petrochemical industry is based (light alkenes, aromatics and methanol), can all be obtained from carbon dioxide. In this context, catalysis plays a key role, enabling the transition of the chemical industry to a circular carbon economy.



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2. Carbon Dioxide Capture, Utilization and Storage (CCUS)

Carbon dioxide can be captured from localized emission sources, such as thermal power plants and large industrial plants (cement plants, refineries, chemical industrial plants, steelworks), taking advantage of processing streams with high concentrations (typically 12–15% in the flue gases of a thermal power station), or directly from the atmosphere (direct air capture, DAC). In the latter case, there is a much greater expenditure of energy, since it is necessary to move large amounts of air through the capture modules by means of fans (remember that the concentration of CO₂ in the atmosphere is only 0.04%).

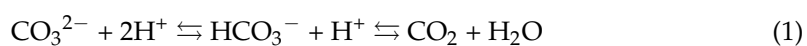
The main techniques available for removing CO₂ from gaseous streams are absorption in aqueous solutions of an alkaline compound (ethanolamine, potassium carbonate, ammonia) and adsorption on porous materials (Table 1). In either case, when saturation is reached, it is necessary to regenerate the absorbent solution or the solid adsorbent, which implies additional energy consumption. The regenerative calcium cycle (or calcium looping) is already a relatively mature technology that allows CO₂ removal from combustion flue gases by reaction with lime (CaO), the resulting calcium carbonate being subsequently calcined to regenerate and reuse the lime. Among the other technologies under development for application in combustion processes are chemical looping combustion (CLC) and Oxyfuel. In both cases, the objective is to obtain a gas stream of pure CO₂ to avoid subsequent separation costs. In CLC, the oxygen needed for the combustion is supplied by means of a metal oxide, which is then reoxidized in a separate reactor and reused in a closed loop; in the Oxyfuel process, oxygen is used for the combustion, instead of air. In addition, a number of promising technologies have emerged, as shown in Table 1 [5].

Table 1. Technologies available for carbon dioxide capture [5].

CO ₂ Capture Technology	Conditions
Absorption	Aqueous solutions of alkanolamines, NH ₃ or K ₂ CO ₃
Adsorption	Solid porous adsorbents
Calcium looping	CaO + CO ₂ ⇌ CaCO ₃ in two fluidized bed reactors
Chemical looping combustion	Metal oxide used to supply oxygen for the combustion
Oxyfuel	Oxygen used for the combustion (instead of air)
Emerging technologies	Membrane separation, ionic liquids, clathrate hydrates

DAC technology could become an important option for achieving carbon neutrality, especially if its energy costs can be reduced [6]. Climeworks (in Switzerland) was the pioneering company in this sector, having started its first commercial unit (with an annual capacity of 900 tons of CO₂) in 2017. A larger plant (“Orca”, 4000 t/year) began operating in Iceland in 2021, injecting the carbon dioxide underground where it is converted into carbonates and stored permanently. This installation uses renewable energy (geothermal). There are other companies developing DAC technology, namely Carbon Engineering, in Canada, and Global Thermostat, in the United States.

On the other hand, the oceans are the largest natural reservoir of anthropogenic CO₂; its average concentration at the surface is around 90 mg/L, two orders of magnitude higher than the concentration in the atmosphere (420 ppm ≈ 0.7 mg/L). At the pH values of seawater (8.1–8.3), carbon dioxide is mainly found in the form of bicarbonate ions (more than 85%), the remaining being in the form of carbonate and dissolved gas, according to the following equilibria:



These equilibria shift to the right as the pH decreases; thus, at $\text{pH} < 6$, most of the carbon is in the form of dissolved CO_2 and can be extracted from seawater.

The first study to assess the feasibility of producing fuels from carbon dioxide dissolved in the ocean dates back to 2010, although at that time there was no technology available for this [7]. The first proposal appeared two years later and was based on a process of electrodialysis of seawater using bipolar membranes, which allow the H^+ and OH^- ions to be separated, originating an acid current (from which CO_2 can be extracted) and an alkaline current [8]. A more interesting proposal is based on the continuous electrodeionization of seawater using a cation exchange module, with simultaneous production of CO_2 and H_2 [9]. Very recently, it was reported that Captura Corporation, a Caltech spin-off, has been set up with the aim of implementing off-shore CO_2 capture using electrodialysis technology. The extraction of CO_2 dissolved in the ocean has an additional advantage, since for every molecule of CO_2 removed, an OH^- ion is released, which promotes the capture of CO_2 from the atmosphere (a process that takes place slowly over months) in order to rebalance the natural carbon cycle [10].

CCUS technologies aim to remove CO_2 from the atmosphere and store it permanently underground in geological formations or use it as a valuable resource, namely as a fluid with technological applications (fire extinguishers, solvents for supercritical extraction, refrigerants, carbonated drinks, tertiary oil recovery) or as a raw material for the chemical industry. The latter application will be developed in the following section.

3. Conversion of Carbon Dioxide into Chemicals and Fuels

Carbon dioxide has been used as a raw material in the chemical industry since the 20th century, mainly in the synthesis of urea (by reaction with ammonia) but also as an additive in the synthesis of methanol and, to a lesser extent, to produce carbonates and inorganic pigments. These applications consume around 220 Mt/year. However, there is a wide range of new processes that make it possible to recycle carbon while mitigating or eliminating the environmentally harmful effects of CO_2 , such as the production of synthetic fuels and high-value-added chemical products [11].

CO_2 conversion reactions can be classified into three groups according to the degree of reduction required and the replacement of C–O bonds by C–H, C–C, C–N or other bonds, as shown in Table 2 [12].

Table 2. Classification of carbon dioxide conversion reactions into chemical products, according to the oxidation state of the carbon atom [12].

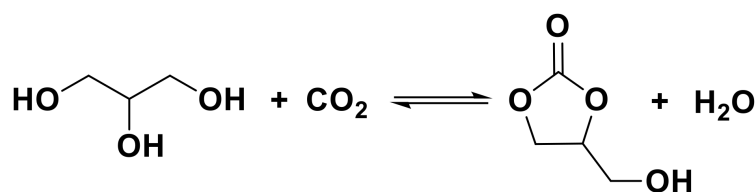
Type of Transformation	Carbon Oxidation State	Chemicals and Fuels
Incorporation	+4	Cyclic carbonates, polycarbonates
Synthesis of functionalized compounds	intermediate	C1 (formic acid, formaldehyde, methanol)
		C2+ (alcohols, acids, aldehydes)
Complete reduction	−4	Methane, >C1 hydrocarbons

At one extreme, there are processes in which the entire CO_2 molecule is incorporated without any change in the oxidation state of the carbon atom, which remains equal to +4. At the other extreme, we have the complete reduction of CO_2 , obtaining saturated hydrocarbons after removing the two oxygen atoms. Examples include the Sabatier reaction (methanation) and the Fischer–Tropsch process. The intermediate group includes processes for the synthesis of functionalized molecules, in which there is both a partial reduction of carbon and the formation of new bonds.

3.1. Synthesis of Carbonates from CO₂

The production of organic carbonates from CO₂ is expected to be a promising technique for the utilization of CO₂, due to the possible synthesis of several chemicals used as electrolytes in Li-Na batteries, as additives for cosmetics, and in coatings, paints and other applications [13]. The most common and effective route to prepare cyclic carbonates, including at the industrial level, is the cycloaddition of CO₂ to epoxides, typically to produce propylene carbonate and ethylene carbonate from propylene and ethylene oxides, respectively, which are two epoxides produced from petroleum-based propylene and ethylene [14]. This is a 100% atom-efficient reaction; however, the nonrenewable origin of the epoxides, along with the high toxicity and volatility of propylene oxide and, in particular, ethylene oxide (which is a gas at room temperature), has motivated the search for other feedstocks to improve the sustainability of the overall process. Alternative routes are being investigated using bio-based wastes, like glycerol, which is a renewable resource available on a large scale as the main side product of the manufacture of biodiesel. Glycerol and CO₂ can be used to produce glycerol carbonate, a valuable chemical that has received much attention over the last several years [15–17] due to its application as a building block in polymer synthesis, beauty/personal care/pharmaceutical products and lithium batteries [18]. Indirect synthetic routes for glycerol carbonate production include the transcarbonation of glycerol with phosgene, carbonation with urea and transesterification with dimethyl carbonate [19]. The most desirable route for glycerol carbonate synthesis is the direct reaction of glycerol with CO₂; however, the conversion is limited by thermodynamics, and finding a suitable dehydrating agent to remove H₂O as the reaction proceeds is essential to reach high yields [20].

In recent years, there have been an increasing number of reports on heterogeneous catalysts used in the synthesis of glycerol carbonate from glycerol and CO₂ (Scheme 1). Some examples of the catalysts that have been reported for this process are compiled in Table 3. Sn catalysts were the first reported catalysts in the carbonation of glycerol with CO₂, followed by several homogenous and heterogeneous catalysts (2-Cyanopyridine; CH₂Br₂; several oxides of Ca, Zn, Sn, Fe, Ce and La; mixtures of La₂O₂CO₃/ZnO; Ti-SBA-15; and Ag-, Zn-impregnated zeolites) [18,21–28]. The approaches developed to greatly improve the glycerol carbonate yield (between 7 and 90%) include the use of organic solvents such as N,N-dimethyl formamide, methanol or 2-cyanopyridine and uneasily recoverable or homogeneous catalysts, high pressures (50–150 bar) and long heating times at high temperatures (80–180 °C). More recently, the addition of Cu nanoparticles (NPs) to a La₂O₃ system allowed up to 33% glycerol conversion, with 45.4% selectivity towards glycerol carbonate (150 °C; 70 bar); the activity was strictly ascribed to the small size of the Cu-NPs and to the presence of basic sites on the support, which could both be involved in the activation of glycerol and CO₂ [29]. Also, CeO₂ nanorods with abundant basic sites and oxygen vacancies showed good glycerol carbonate yields (up to 78.9%) under conditions of 150 °C and 40 bar in 5 h [25]. Investigations have been carried out using a few carbon-based catalysts for the coupling of CO₂ with epoxides but mainly in direct glycerol carbonate synthesis [30–32]. CO₂ electro-cycloaddition to epoxides has also been investigated under mild electrochemical conditions for the synthesis of cyclic carbonates [33–35].



Scheme 1. Glycerol carbonylation with CO₂ to form glycerol carbonate. Reprinted from reference [36].

Table 3. Examples of homogenous and heterogenous catalysts tested in the direct synthesis of glycerol carbonate from CO₂.

Catalyst	Temperature (°C)/ Pressure (Bar)	Solvent/Dehydrating Agent	GC Yield (%)	Ref.
n-Bu ₂ Sn(OMe) ₂	180/50	-/Molecular sieves	5.72	[37]
nBu ₂ SnO	120/138	MeOH/13X (soda) zeolite	35	[38]
Zn(OTf) ₂ /phen	180/80	N-methyl-2-pyrrolidone/CaC ₂	92	[39]
ZnY	180/100	-/-	5.8	[28]
ZnO	180/150	-/-	8	[23]
La ₂ O ₂ CO ₃ /ZnO	170/40	CH ₃ CN	14.4	[26]
Cu/La ₂ O ₃	150/70	CH ₃ CN	45.4	[40]
Cu/MgO Cu/La ₂ O ₃	150/40	CH ₃ CN	26.1 29.3	[41]
CeO ₂ nanorods	150/40	DMF/2-cyanopyridine	78.9	[25]
CeO ₂ nanopolyhedra	180/150	2-cyanopyridine	14.2	[21]
MgO	150/80	DMF/2-cyanopyridine	10.6	[42]
LaCoO ₃	150/30	DMF/2-cyanopyridine	72.4	[43]

GC—glycerol carbonate; MeOH—methanol; TEGDME—tetra(ethylene glycol)dimethyl ether; DMF—dimethyl formamide.

3.2. Conversion of CO₂ into C1 and C2+ Chemicals and Fuels

The processes for converting carbon dioxide with “green hydrogen” (obtained by electrolysis of water with electricity from renewable sources) are generically known as “power-to-X”; examples include “power-to-gas” (methane production), “power-to-liquids” (liquid hydrocarbon production) and “power-to-chemicals” (chemical synthesis), as shown in Table 4.

Table 4. Production of fuels and chemicals from carbon dioxide and green hydrogen: “power-to-X” processes. Sources: references [11,12,44].

Power-to-X	Products Obtained and Processes Involved
Power-to-gas	Methane: Sabatier reaction (CO ₂ methanation)
Power-to-liquids	Liquid hydrocarbons: “reverse water–gas shift” (RWGS), followed by Fischer–Tropsch synthesis
Power-to-chemicals	Synthesis of chemicals by catalytic hydrogenation or by electrocatalytic reduction

The catalytic hydrogenation of CO₂ to produce methane is the “Sabatier reaction”:

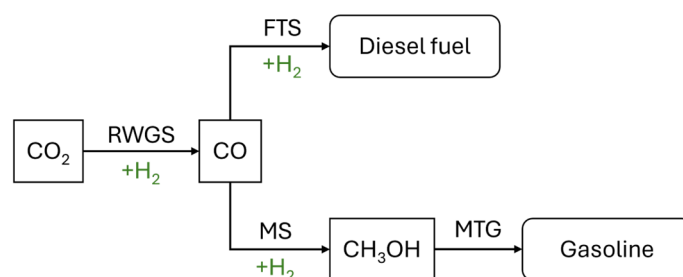


Sabatier and Senderens studied this reaction at the beginning of the 20th century, using Ni catalysts [45]. It is now known that all metals in groups 8–10 of the Periodic Table can catalyze the reaction, Ru being the most active and Ni the most selective [46]. On the basis of cost-effectiveness, nickel is also the catalyst of choice.

The “power-to-gas” process involves converting solar energy into electricity (efficiency = 20%), producing hydrogen by water electrolysis (efficiency = 90%) and methanating carbon dioxide (efficiency = 70%). This results in an overall process efficiency of

12.6%, which is already better than what plants can achieve (1.8–2.2%), or even microalgae (6–10%) [11].

Traditionally, the production of synthetic fuels is based on synthesis gas ($\text{CO} + \text{H}_2$), which can be obtained from fossil raw materials (coal, oil or natural gas) by reacting them with water vapor and/or oxygen: steam-reforming, partial oxidation and gasification processes. Synthesis gas can be used to produce methanol (methanol synthesis) or liquid hydrocarbons (Fischer–Tropsch synthesis). The “power-to-liquids” process is already being implemented at an industrial level using the existing technology, which involves the initial conversion of CO_2 to CO (via the RWGS reaction), followed by the conventional Fischer–Tropsch process (FTS). This is the first of the indirect routes shown in Scheme 2. Catalysts for the RWGS reaction are usually based on iron oxides or on copper, but there is intense research on the development of new catalysts [47–50]. cobalt and iron are the most widely used FTS catalysts to produce diesel fuels [51].



Scheme 2. Indirect routes to produce liquid fuels from carbon dioxide (“power-to-liquids”), with green hydrogen.

Liquid hydrocarbons can also be produced by the methanol-to-gasoline (MTG) process developed by Mobil in the 1970s. In this alternative indirect route, CO_2 is first converted to CO by the RWGS reaction, followed by methanol synthesis (MS); then, the MTG process converts methanol into high-octane gasoline on a shape-selective zeolite catalyst (ZSM-5, MFI structure). The mechanism of the reaction proceeds through a sequence of steps involving the initial dehydration of methanol into dimethyl ether (DME) and the formation of light alkenes (C_2 – C_5) and C_5 + alkenes, then alkanes, cycloalkanes and aromatics. The final product is a mixture of hydrocarbons in the gasoline boiling range (C_5 – C_{10}) [52–55]. Methanol-to-olefins (MTO) and methanol-to-aromatics (MTO) are related processes [56].

Obviously, these indirect routes are not an ideal solution, since the RWGS reaction consumes valuable “green” hydrogen to produce water:



The same applies to the methanation reaction, where 50% of the hydrogen ends up in water.

The direct use of carbon dioxide in “power-to-liquids” processes is possible but requires some modifications, as discussed in Sections 3.2.1 and 3.2.2.

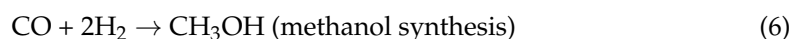
Concerning the synthesis of chemicals (“power-to-chemicals”), in addition to C_1 compounds (methanol, formic acid, formaldehyde) and their derivatives, the greatest research efforts are focused on the synthesis of oxygenated C_2 + compounds (acids, alcohols, aldehydes, ethers) [12,44,57], as well as light olefins (C_2 – C_4), straight-chain α -olefins (C_4 – C_{17}) and aromatic hydrocarbons (C_6 – C_8) [58]. The formation of C_2 + products requires not only efficient CO_2 activation and reduction but also selective carbon–carbon (C – C) coupling, which is a significantly more challenging process [59–62]. The “one-pot” hydrogenation of CO_2 via the FTS has attracted significant attention due to its operational simplicity—stemming from a reduced number of process steps—and consequently lower associated

costs and energy consumption [63]. This integrated process is particularly challenging, as it requires bifunctional catalysts with active sites capable of catalyzing both the RWGS (CO generation) and FTS (CO hydrogenation) steps under identical operating conditions [64], although the RWGS reaction is mildly endothermic and competes with the strongly exothermic CO₂ methanation reaction at lower temperatures [49]. Supported catalysts based on Fe, Co and Ru are among the most widely used, with Fe being particularly effective for olefin production [65–68] and noble metals like Ru/Pd showing higher activity toward alcohol formation [69,70].

Because the pathway involved in the growth of the carbon chain for any given C₂+ product contains many successive reaction steps [71], tandem catalysis, which involves separate but coupled steps (e.g., CO₂ to CO followed by CO coupling), is gaining attention for modular and efficient C₂+ production [72]. Recent research has focused intensively on the development of efficient and selective bifunctional catalysts.

3.2.1. Hydrogenation of Carbon Dioxide to Methanol

In the synthesis of methanol, a copper metal and zinc oxide catalyst supported on alumina is currently used (a “low-pressure” process at 50–100 bar and 220–300 °C, introduced by ICI in 1966, replacing the original “high-pressure” process developed by BASF in the 1920s). The Cu-ZnO/Al₂O₃ catalyst also works with a CO₂ + H₂ mixture; in fact, in the conventional synthesis process, it is already usual practice to add some CO₂ (2–8%) to the synthesis gas, to adjust the stoichiometry and to improve the kinetics of the reaction [73]. The methanol is actually produced from CO₂ (through the formation of an intermediate formate species adsorbed on the catalyst) according to the following mechanism, where O_a is an adsorbed oxygen atom [74]:



However, the hydrogenation of carbon dioxide produces not only methanol but also water:



Water can also be formed by the reverse water–gas shift (RWGS) reaction.

Although the conventional catalyst works well for CO₂ hydrogenation, the formation of water can affect its stability, so it is necessary to develop new catalysts that are more tolerant to the presence of water [73,75]. A Cu-ZnO/ZrO₂ catalyst was proposed as a promising alternative [76]. More recently, In₂O₃/ZrO₂ was reported as a very active and stable catalyst for the synthesis of methanol from CO₂, with 100% selectivity [77]. A further improvement was achieved by promotion with Pd, which is a good catalyst for hydrogen dissociation [78]. A subsequent investigation of this catalyst system by operando spectroscopy showed that the two sites perform separate functions: Pd activates H₂, while In₂O₃ activates carbon dioxide [79]. Another option would be the direct synthesis of dimethyl ether (DME) from carbon dioxide, using bifunctional catalysts, by coupling a MS catalyst with a dehydration component [80]. Core–shell catalysts, where the MS component is in the core and the dehydration component is in the shell, have shown enhanced performance for DME synthesis from syngas [81]. The application of this type of nanostructured catalyst for CO₂ hydrogenation has recently been reviewed [82].

Methanol is a very versatile compound as it can be used as an energy carrier and as a fuel (directly or in fuel cells) or as a base chemical for the production of a wide variety of

organic compounds: formaldehyde, acetic acid, ethers (MTBE, TAME, DME), methylamines, methyl chloride, methylmethacrylate, dimethylterephthalate, olefins (MTO process) and aromatics (MTA process). It can also be converted into gasoline (MTG process) [12]. The MTG, MTO and MTA processes are based on the use of the ZSM-5 zeolite developed in the 1970s by Mobil [83]. This versatility of methanol led George Olah to propose the "methanol economy" as an alternative to the "hydrogen economy" [84], since methanol has obvious advantages as an energy carrier: it is a liquid at room temperature, which makes it easier to transport and store; and its energy density (15.6 MJ/L), although only about half that of traditional fuels (38.6 MJ/L for diesel, 34.2 MJ/L for gasoline), is much higher than that of hydrogen (8.6 MJ/L in the liquid state, under cryogenic conditions) [73].

3.2.2. Multifunctional Catalysts for Tandem Conversion of CO₂ to Hydrocarbons

The direct hydrogenation of carbon dioxide to liquid fuels (gasoline and diesel) can be achieved via methanol or the FTS, with multifunctional catalysts [63]. Gao et al. [71] reported the direct conversion of CO₂ to gasoline-range hydrocarbons using a bifunctional catalyst consisting of In₂O₃ and HZSM-5 (mass ratio 2:1). The selectivity to C₅–C₁₁ hydrocarbons was 78.6%, mostly iso-alkanes (59.2%) and aromatics (14.6%), which are high-octane-number hydrocarbons. On the other hand, the CH₄ selectivity was only 1% and the CO₂ conversion rate was 13.1%. The unwanted RWGS reaction was suppressed as a result of the proximity of the two catalyst components, In₂O₃ for methanol synthesis and HZSM-5 for C–C coupling. Moreover, different products were obtained by replacing ZSM-5 with other zeolites: 76.9% selectivity to lower alkenes (C₂–C₄) with SAPO-34 and 38.7% selectivity to LPG (C₃–C₄ alkanes) with zeolite beta. The direct hydrogenation of CO₂ to gasoline-range hydrocarbons was also reported by Wei et al. using a Na-promoted Fe₃O₄/HZSM-5 catalyst [61]. Iron oxide provides the active sites for the RWGS and FTS reactions. When the catalyst is reduced with H₂ before the reaction, metallic iron is formed, which is converted into Fe₃O₄ and Fe₅C₂ by the reaction mixture. The Na promoter increases the surface basicity, facilitating the carburization of the iron catalyst. Iron carbides (in this case, Fe₅C₂) are the active phases involved in the FTS [85,86]. Thus, CO₂ is initially reduced to CO on Fe₃O₄, which is the active phase for the RWGS; then, CO is hydrogenated to α -alkenes on Fe₅C₂, which are subsequently converted to C₅–C₁₁ hydrocarbons on the acid sites of the zeolite HZSM-5. This multifunctional catalyst promotes the tandem conversion of CO₂ into gasoline with a selectivity of 78% and only 4% methane selectivity at a CO₂ conversion rate of 22%. By replacing HZSM-5 with the zeolite HMCM-22, higher yields of iso-alkanes and lower yields of aromatics are obtained, which is a more favorable composition for gasoline. However, coke formation increases, leading to catalyst deactivation [87].

The hydrogenation of CO₂ to α -olefins and aromatics has recently been reviewed [88]. The α -olefins with a carbon chain of length ≥ 4 are used for the synthesis of polymers, surfactants, lubricants and plasticizers; they are currently produced from oil-derived chemicals. Light olefins (ethene, propene, butenes) and aromatics (benzene, toluene, xylenes), together with methanol, are the so-called "base chemicals" (or primary products) of the petrochemical industry, from which all other products are obtained. The main routes to obtain olefins are the CO₂-FTS route, combining the RWGS reaction with subsequent hydrogenation of CO to olefins, and the CO₂-MS route, including CO₂ hydrogenation to methanol and the MTO process. Iron-based catalysts with promoters are used in the first case, while the methanol route involves oxide catalysts coupled with zeolites. Aromatics can also be obtained by a CO₂-FTS route, or by a CO₂-MS route. In the CO₂-FTS route, iron-based catalysts with different promoters and supports are coupled with an acidic zeolite, usually HZSM-5. For the CO₂-MS route, a bifunctional catalyst coupling the MS and MTA

reactions is required. Since the zeolites used for MTA usually operate at temperatures above 400 °C, MS catalysts based on Cu cannot be used, as they are not stable above 300 °C. So, high-temperature MS catalysts (based on Zn, Zr, In or Cr oxides) are coupled with the HZSM-5 zeolite to perform this tandem reaction. For instance, a bifunctional ZnZrO/HZSM-5 catalyst was reported to convert CO₂ to aromatics with high selectivity (73% at a conversion rate of 14%). The presence of water (formed on the MS catalyst) and CO₂ was found to inhibit the formation of polycyclic aromatics (which are precursors for coke formation), leading to excellent stability [89].

A comparison of these two routes for aromatics shows that the CO₂ conversion rate in the FTS route is usually higher, and the CO selectivity is lower; thus, it is more efficient than the MS route, although it suffers from a wider product distribution and higher methane production. The main challenge in the MS route is to increase CO₂ conversion while preventing the formation of CO from the RWGS reaction. The product distribution can be adjusted by tuning the acid strength and pore size of the zeolites [88,90].

3.2.3. Electrocatalysis

The electrochemical CO₂ reduction reaction (CO₂RR) is also an appealing idea for CO₂ abatement, allowing the direct conversion of carbon dioxide into species that are more reduced (CO, formic acid, methanol, alcohols and C₂+ hydrocarbons) [91–93]. The industrial sector is very interested, with some companies being created to (electro)recycle CO₂ into chemicals and fuels (OPUS12, Berkeley, CA, USA) or leading big projects on the theme (Siemens and Evonik, Solvay) [94]. However, most of the electrochemical conversion technologies remain at low technology readiness levels (TRLs) and need further development before full-scale deployment [95]. To push this electrochemical technology forward, the development of advanced CO₂ reduction electrocatalysts is crucial. Considerable progress has already been achieved in CO₂RR to one-carbon (C₁) products (such as CO and formic acid), with high Faradaic efficiency (>95% selectivity) for, e.g., CO being reported [96]. In contrast, CO₂RR into more-added-value C₂+ chemicals (such as ethanol, ethylene and propanol) is still a challenging task; monometallic electrocatalysts cannot efficiently reduce CO₂ to these products [97], with low selectivity, competitive hydrogen evolution reactions (HERs) and poor stability being reported [98]. Transition metal (Mo, Fe, Co and Ni) [99] catalysts associated with conductive or semiconductive materials (cheap metal oxides or carbon derivatives, like carbon nanotubes [100,101]), are a promising way to obtain higher efficiencies [102], more so when heteroatom dopants (N, B, P, S) are used to adjust the carbon material's surface chemistry [103–106]. For example, N-doped graphene quantum dots revealed 90% Faradaic efficiency (FE) in CO₂ reduction to C₂H₄ and 45% in C₂H₅OH formation. N-doped nanodiamond/Si rod arrays have been reported to reach FEs from 91.2 to 91.8% at −0.8 to −1.0 V vs. RHE in 0.5 M NaHCO₃ for CO₂ to acetate. N-doped ordered mesoporous carbon containing pyridinic and pyrrolic N species was also reported to have superior performance towards C₂H₅OH (77% FE at −0.56 V vs. RHE was achieved) [107]. Besides N doping, the introduction of boron, sulfur, and fluorine atoms has been attempted, but most reports deal with C₁ product formation [103].

Another possibility is “artificial photosynthesis”, based on the conversion of solar energy into chemicals, in which water is the source of protons and electrons and the reactions of carbon dioxide reduction and water oxidation are combined in the presence of suitable catalysts. These processes are still at the research stage, but their potential contribution to decarbonizing the chemical industry is enormous [108]. Approaches may include photothermal [109–112] and photoelectro [112–116] CO₂ catalytic conversion.

4. Summary and Perspectives

The chemical industry has already begun to adapt to the new challenges of sustainability, which involve replacing oil, coal and natural gas with renewable raw materials. But the concept of the “circular economy” applied to the carbon used by the chemical industry necessarily implies the use of carbon dioxide as a raw material. Nature teaches us that CO₂ is renewable carbon, and, as such, it must be captured from localized sources, taken directly from the atmosphere, or extracted from the oceans and converted into fuels and chemicals. The production of methane (by the Sabatier reaction) is already carried out with reasonable energy efficiency using solar energy, while methanol and liquid hydrocarbons can be obtained from CO₂ by means of existing technologies (originally developed for the conversion of synthesis gas) using “green hydrogen”; however, these processes can still be optimized, which requires the development of new catalysts. On the other hand, the synthesis of chemicals, particularly C₂+ compounds, by catalytic hydrogenation or electrocatalytic reduction of carbon dioxide, is an area that has been the subject of intense research activity. We have particularly focused on the synthesis of hydrocarbons, namely light alkenes and aromatics, by direct hydrogenation of carbon dioxide in tandem reactions with multifunctional catalysts. Together with methanol, these hydrocarbons are the “base chemicals” on which the petrochemical industry is based. Promising results have already been achieved by improving the catalysts for methanol and Fischer–Tropsch syntheses, coupled with shape-selective zeolites. These processes still require optimization in order to enhance the activation of CO₂ and improve the selectivity to the desired products and the stability of the catalysts.

Meanwhile, photocatalytic and photoelectrocatalytic CO₂ conversion processes, which are still at a less advanced stage of development, offer the possibility of eventually achieving the ultimate goal of “artificial photosynthesis”, mimicking nature. In all these processes, catalysis emerges as a key technology for the efficient conversion of CO₂, promoting the chemical industry’s transition to a circular carbon economy, reducing resource consumption and waste production, and minimizing the environmental impacts of its activity.

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